Jark-Wah Andrew Wong

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EDUCATION

The Pennsylvania State University — **State College, PA**
Doctor of Philosophy in Chemical Engineering — Aug. 2019 - Present
Doctoral Minor in Computational Materials — **Anticipated Graduation – July 2024**
Thesis: Deconvoluting electrocatalysis using DFT and Classical MD — GPA: 3.87/4.0
Advisor: Dr. Michael J. Janik

Texas A&M University — **College Station, TX**
Bachelor of Science in Chemical Engineering, *magna cum laude* — Aug. 2015 – May 2019
Certification: Engineering Therapeutics Manufacturing — GPA: 3.7/4.0
Dean's List (2016-2019)
Advisor: Dr. Hung-Jen Wu

RESEARCH EXPERIENCE

Computational Chemistry and Materials Science Summer Institute Internship
Lawrence Livermore National Laboratory, Dr. Sneha Akhade — May 2023 – Present
Collaboration: Dr. Christopher Hahn of Lawrence Livermore National Laboratory

- Guiding computational efforts between multi-disciplinary group for biomass upgrading on effects of metal support interactions on biomass oxidation electrochemistry
- Utilizing DFT methods to determine electrocatalyst design principles for the electro-oxidation of HMF and alcohol in biomass across Ni oxide surfaces supported on Au or Ag underlayers

Understanding the dependence between the activity of CO_2 electro-reduction, identity of alkali metal cations, and surface facets of Au within electrode-electrolyte interface
The Pennsylvania State University, Dr. Michael J. Janik — Mar. 2023 – Present
Collaboration: Dr. Anne Co of Ohio State University

- Leading computational efforts using DFT/FF-MD methods to better understand the impacts of the electrolyte's ions and metal facet dependence during CO_2 electro-reduction on Au surfaces
- Studying changes in EDL interfacial properties due to different electrolyte compositions and surface facets using large scale FF-MD simulations

Electro-oxidative valorization of biomass: design strategies for selective and stable catalysis
The Pennsylvania State University, Dr. Michael J. Janik — Nov. 2022 – Present
Collaboration: Dr. Adam Holewinski of University of Colorado at Boulder

- Determining fundamental activity and selectivity descriptors for the design of oxidation bimetallic electrocatalysts for biomass upgrading of multi-carbon organic molecules using DFT methods
- Deciphering vibrational frequencies of furfural intermediate adsorbed on Cu from experimentally obtained ATR-SEIRAS spectra using DFT methods

DFT and FF-MD methods to study electrochemical double-layers and electrode-electrolyte effects
The Pennsylvania State University, Dr. Michael J. Janik — Jan. 2020 – Present
Collaboration: Dr. Scott T. Milner of The Pennsylvania State University

- Developing a combined DFT/FF-MDMD approach to model electrochemical double layer structures and its effects on electrocatalytic barriers, adsorption, and electrocatalytic reaction energetics
- Probing specific alkali cation adsorption across late transition metal surfaces and sensitivities of modeling electrode-electrolyte effects during adsorption using DFT and Classical MD methods

Deconvolution of charge-transfer, mass transfer, and ohmic resistance in phosphonic acid-sulfonic acid ionomer binders in electrochemical hydrogen pumps

The Pennsylvania State University, Dr. Michael J. Janik Feb. 2023 – Present

Collaboration: Dr. Chris Arges of Penn State University

- Studying effects of adsorption affinity of phosphonic acid ionomers on charge resistance values during HER/HOR kinetics in electrochemical hydrogen pumps using DFT methods

Investigating the electrocatalytic reduction mechanism of nitroaromatics using DFT methods

The Pennsylvania State University, Dr. Michael J. Janik Jan. 2020 – May 2022

Collaboration: U.S. Army, Catalyze H_2O, and Dr. Lauren Greenlee of The Pennsylvania State University

- Evaluated and screened potential TNT and nitrobenzene reduction electrocatalysts across late transition metal and metal oxide surfaces for the environmental treatment of wastewater using DFT methods

Developing nanoplasmonic sensors coupled with Raman spectroscopy for detection applications

Texas A&M University, Dr. Hung-Jen Wu Dec. 2017 – May 2019

- Experimentally synthesized Ag-based catalysts through vapor deposition and Tollens' test
- Characterized Ag deposited catalysts using Raman spectroscopy for drug detection applications

Evaluating the impact of prenatal stress on the endocrine system

Texas A&M University, Dr. Thomas H. Welsh Jr. Jan. 2017 – May 2017

- Evaluated large experimental data sets and developed trends using MATLAB for stress research

RESEARCH PUBLICATIONS

Peer-review papers

Z. Cui, **A. J. Wong,** D. R. Zhu, M. J. Janik, and A. Co. "Understanding cation effect of CO_2 reduction on single-crystal and polycrystalline gold electrodes under well-defined mass transport conditions. Manuscript in Preparation.

K. Arunagiri, **A. J. Wong**, L. Briceno-Mena, J. A. Romagnoli, M. J. Janik, and C. G. Arges. *"Deconvoluting charge-transfer, mass transfer, ohmic resistances in phosphonic acid-sulfonic acid ionomer binders used in electrochemical hydrogen pumps".* Submitted to Advanced Energy Materials.

A. J. Wong, B. H. Tran, N. Agrawal, D. R. Zhu, S. T. Milner, and M. J. Janik. "*Modeling Electrocatalysis: Best practices for incorporating electrode-electrolyte effects* ". Manuscript in preparation.

A. J. Wong, B. H. Tran, S. T. Milner, and M. J. Janik. "*Incorporating electrode-electrolyte interfacial effects during specific cation adsorption on late transition metals using Density Functional Theory methods*". Manuscript in preparation.

A. J. Wong, J. L. Miller, B. Perdue, and M. J. Janik. "*Investigating the electrocatalytic reduction of 2,4,6-tri-nitro-toluene (TNT) using Density Functional Theory methods*". **Green Chem.** Advanced Article, May 19th, 2023

A. J. Wong, J. L. Miller, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory*". **Cell Chem Catalysis**, Vol 2, Issue 6, P1362-1369, June 16, 2022

N. Agrawal, **A. J. Wong**, S. Maheshwari, and M. J. Janik. "An efficient approach to compartmentalize double layer effects on kinetics of interfacial proton-electron transfer reactions". Submitted to Journal of Catalysis.

J. L. Miller, **A. J. Wong**, and M. J. Janik. "*Electrocatalytic properties of an iron organometallic complex and its role in the reduction of nitroaromatic compounds*". Manuscript in preparation.

Book Chapters

N. Agrawal, K. Yeh, S. Maheshwari, B. Tran. **A. J. Wong**, and M. J. Janik. *"Density Functional Theory Methods for Electrocatalysis."* Book chapter in submission.

Oral Presentations

AIChE Annual Meeting, Orlando, Nov. 2023. **A. J. Wong**, B. Tran, N. Agrawal, D. R. Zhu, S. T. Milner, and M. J. Janik. *"Multi-Scale Modeling of Electrocatalytic Processes within the Electrochemical Double Layer."*

NAM 28th Annual Meeting, Rhode Island, June 2023. **A. J. Wong,** B. Tran, S. T. Milner, and M. J. Janik. *"Modeling Specific Alkali Cation Adsorption and Electrochemical Double-Layers with DFT and Classical MD."*

AIChE Annual Meeting, Phoenix, Nov. 2022. **A. J. Wong**, B. Tran N. Agrawal, S. T. Milner, and M. J. Janik. "*Incorporating Electrode-Electrolyte Interfacial Effects on the Specific Adsorption of Ions on Late Transition Metal Surfaces Using a Combined DFT/FF-MD Approach*."

AIChE Annual Meeting, Phoenix, Nov. 2022. B. Tran, **A. J. Wong**, N. Agrawal, S. T. Milner, and M. J. Janik. "*A Combined Classical MD and DFT Approach for Modeling Ionic Adsorption on Metal Electrodes with Explicit Description of the Electrical Double Layer*."

PSU ChE Research Symposium, University Park, Sept. 2022. **A. J. Wong,** B. Tran, S. T. Milner, and M. J. Janik. "*Incorporating Electrode-Electrolyte Interfacial Effects on the Specific Adsorption of Ions on Late Transition Metal Surfaces Using a Combined DFT/FF-MD Approach*."

ACS Spring 2022 Meeting, San Diego, March 2022. **A. J. Wong.** J. L. Miller, B. Perdue, and M. J. Janik. *"Elementary mechanisms of the electrocatalytic reduction of nitroaromatics on late transition metal surfaces using Density Functional Theory methods."*

AIChE Annual Meeting, Boston, Nov. 2021. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. *"Mechanistic Studies on the Electrocatalytic Reduction of Nitroaromatic Compounds."*

Poster Presentations

AIChE Annual Meeting, Orlando, Nov. 2023. **A. J. Wong**, B. Tran, M. J. Janik. "Modeling Electrode-Electrolyte Interfacial Effects and Specific Alkali Metal Cation Adsorption Using DFT/FF-MD Approach."

2023 Catalysis Club of Philadelphia Annual Symposium-Student Poster Joint Conference. Philadelphia, May. 2023. **A. J. Wong**, B. H. Tran, D. R. Zhu, S. T. Milner, and M. J. Janik. **"**Modeling Electrode-Electrolyte Interfacial Effects and Specific Alkali Metal Cation Adsorption Using DFT/FF-MD Approach**."**

Pittsburgh-Cleveland Catalysis Society Annual Symposium. Pittsburg, May. 2023. **A. J. Wong**, B. H. Tran, D. R. Zhu, S. T. Milner, and M. J. Janik. **"**Modeling Electrode-Electrolyte Interfacial Effects and Specific Alkali Metal Cation Adsorption Using DFT/FF-MD Approach**."**

AIChE Annual Meeting, Phoenix, Nov. 2022. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Investigating the Electrocatalytic Reduction of 2,4,6-Tri-Nitro-Toluene (TNT) across Late Transition Metal Surfaces Using Density Functional Theory Methods*."

PSU ChE Research Symposium, University Park, Sept. 2022. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory methods."*

Pittsburgh-Cleveland Catalysis Society Annual Symposium. Pittsburg, Aug. 2022. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory methods.*"

AIChE Annual Meeting, Boston, Nov. 2021. J. L. Miller, B. Perdue, **A. J. Wong**, and M. J. Janik. "Electrocatalytic Reduction of Nitrobenzene by Iron-Salen Ligand Complexes"

Pittsburgh-Cleveland Catalysis Society Annual Symposium. Pittsburg, Oct. 2021. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory methods.*"

PSU ChE Research Symposium, University Park, Oct. 2021. **A. J. Wong**, J. L. Miller, B. Perdue, and M. J. Janik. "*Elementary mechanism for the electrocatalytic reduction of nitrobenzene on late transition metal surfaces from Density Functional Theory methods.*"

TEACHING & WORK EXPERIENCE

Graduate Teaching Intern and Assistant | The Pennsylvania State University Jan. 2022 – May 2022
Course: Chemical Reaction Engineering

- Taught and guided over 70 students through a "flipped classroom", where students practice new problems each class and consult the TA for reinforced learning
- Designed and implemented weekly group problems and exam problems to test fundamental reaction engineering knowledge through unique and challenging problem sets
- Organized multiple in-person/remote office hours weekly and monthly review sessions

Hardware and Software Lab Manager | The Pennsylvania State University Jan. 2020 – Present

- Manage Janik lab hardware for efficient research through collaboration with PSU IT department
- Supervise supercomputing allocation usage across clusters for over 12 lab students
- Design and update the Dr. Janik lab website with research updates

Undergraduate Teaching Assistant | Texas A&M University Aug. 2018 – Dec. 2018
Course: Elementary Chemical Engineering

- Managed recitation of over 200 students to teach fundamental mass balance and chemical engineering principles through group practice problems

Private Lesson Teacher | Frisco Independent School District July 2015 – Aug. 2016

- Taught music and leadership principles to individuals or groups of over 50 students
- Managed lesson plans during leadership camp encompassing over 200 students

EXTRACURRICULAR AND LEADERSHIP ACTIVITIES

Temple University Community Service Association Aug. 2022 – Present
Member, Volunteer

- Assist and stocked South Philadelphia community fridges with food and drink for lower-income and minority derived communities in Philadelphia

Penn State Chemical Engineering Graduate Student Association Aug. 2019 – Present
Mentor, Member

- Advised and mentored five first-year graduate student in their Ph.D. career and qualifying exam preparations
- Assisted and participated in the 2021 and 2022 PSU ChE Research Symposium

American Institute of Chemical Engineers Mar. 2017 – Mar. 2019

Vice President, Publicity

- Awarded best national AIChE student chapter during VP tenure
- Coordinated record high meetings/workshop attendances for prospective and current ChE students
- Chaired session at the SPARK Conference to teach middle and high school students about engineering

Omega Chi Epsilon Chemical Engineering Honor Society Dec. 2017 – May 2019

Mentor, Member

- Invited to the organization to mentor engineering students due to outstanding academic achievement
- Assisted in annual Big Event to perform volunteer work for the community in Bryan, TX

HONORS AND AWARDS

- NAM28 Kokes Award – Feb. 2023
- Dean's List – Jan. 2016 to May 2019

TECHNICAL SKILLS

Density Functional Theory, Vienna Ab Initio Simulation Package (VASP), Gaussian16, Python, Atomic Simulation Environment (ASE), Classical Molecular Dynamics, GROMACS, Amsterdam Modeling Suite (AMS), Materials Studio, VESTA, MATLAB, Apache Spark, TensorFlow, Wolfram Mathematica, and Microsoft Office Suite